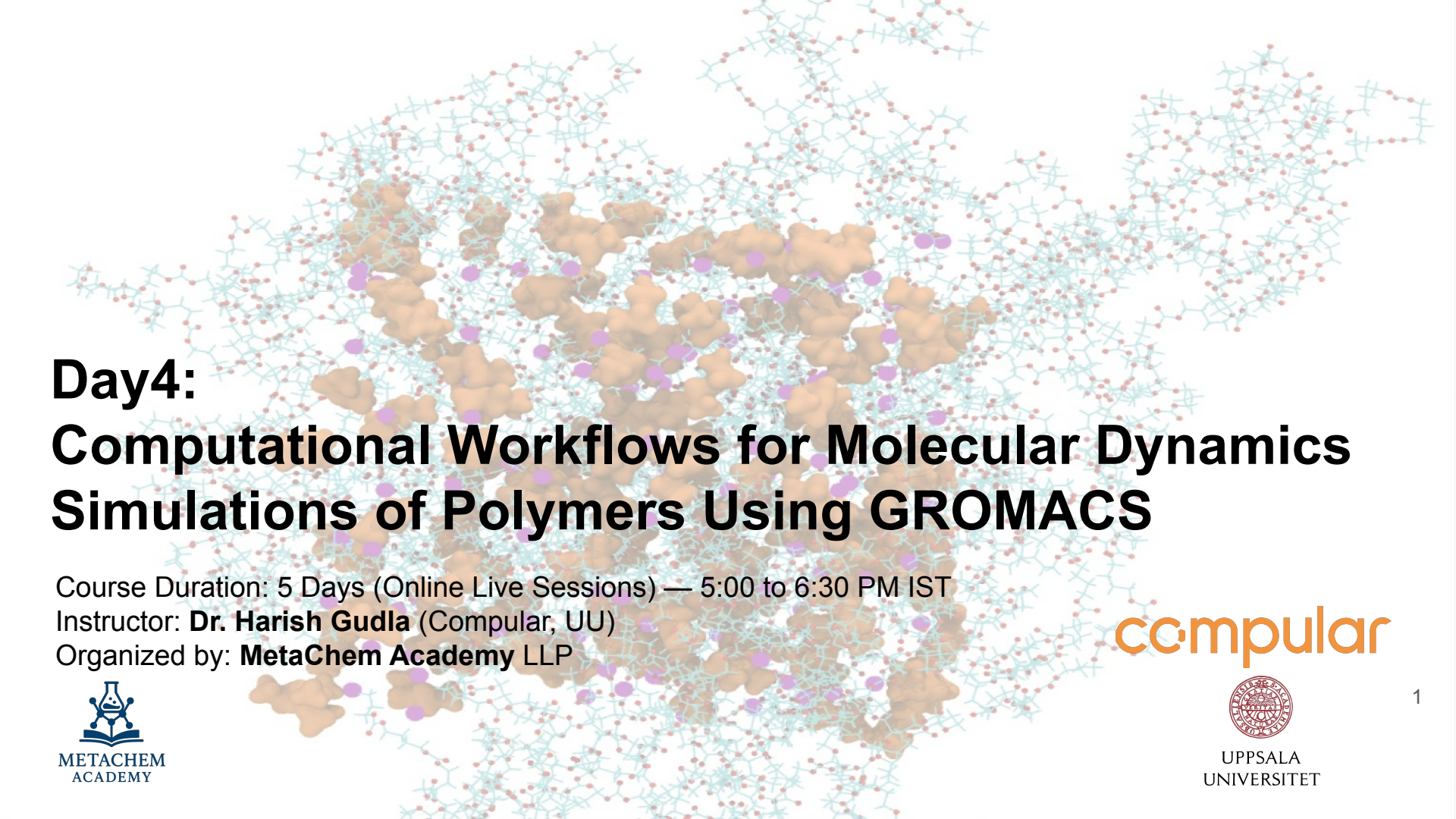




Day4: Computational Workflows for Molecular Dynamics Simulations of Polymers Using GROMACS

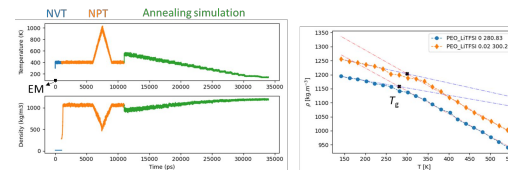
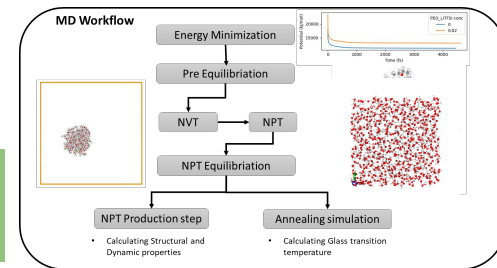
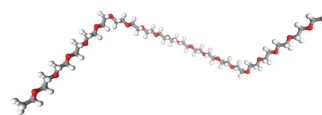
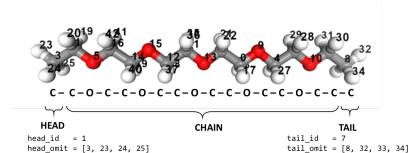
Course Duration: 5 Days (Online Live Sessions) — 5:00 to 6:30 PM IST

Instructor: **Dr. Harish Gudla** (Compular, UU)

Organized by: **MetaChem Academy LLP**

Workshop Schedule

Date	Theory Session (5:00–5:30 PM IST)	Hands-On Session (5:30–6:30 PM IST)
10 Dec	Introduction to classical MD, force fields, and motivation for polymer simulations	Installing and setting up tools and Parameterizing monomers
11 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Getting to jupyter notebooks and Parameterizing monomers Building polymer chains using parameterized monomers and preparing input files
12 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Building polymer chains using parameterized monomers and preparing input files
15 Dec	GROMACS input formats and MD simulation workflows	Run first GROMACS MD on single polymer chain and perform initial analysis
16 Dec	Thermal and conformational analysis: annealing simulations and glass transition temperature (T_g) concepts	Performing annealing runs and calculating polymer properties: density, RMSD, R_g , and T_g



Day 3 - Outline

- Recap
 - Polymer initial files (.gro,.top) using ambertools
- Theory session (5:00-5:30 pm IST)
 - GROMACS input/output formats
 - MD simulation workflows
- Hands-on session (5:30-6:00 pm IST)
 - GROMACS MD on single polymer chain
 - Preliminary Analysis

Classical Force Field

The functional form of a typical forcefield

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{Coul}} + U_{\text{Bond}} + U_{\text{Angle}} + U_{\text{Dihedral}} + U_{\text{Improper}}$$

Non-bonded interactions

$$U_{\text{LJ}} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right],$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_1\epsilon_0 r_{ij}},$$

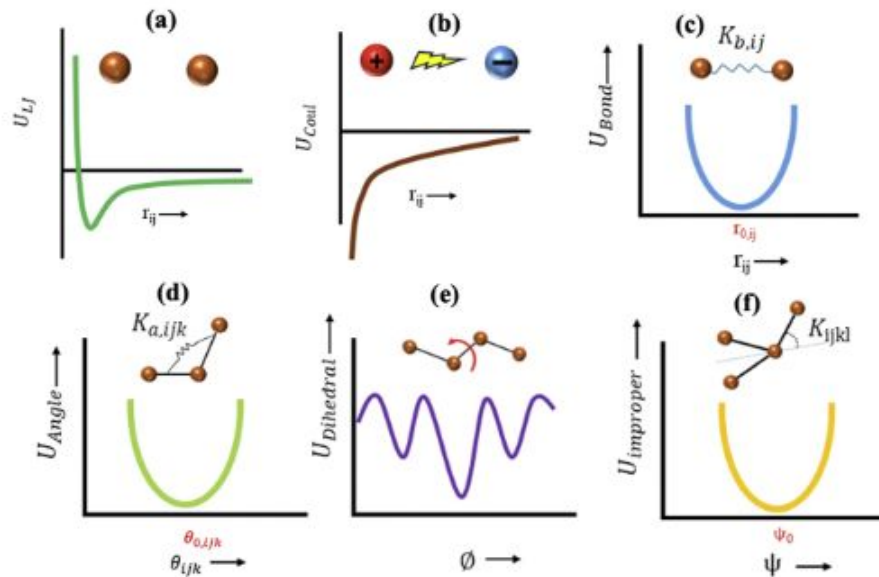
Bonded interactions

$$U_{\text{Bond}} = K_{b,ij} (r_{ij} - r_{0,ij})^2,$$

$$U_{\text{Angle}} = K_{a,ijk} (\theta_{ijk} - \theta_{0,ijk})^2,$$

$$U_{\text{Dihedral}} = \frac{1}{2}K_1[1 + \cos(\phi)] + \frac{1}{2}K_2[1 - \cos(2\phi)] \\ + \frac{1}{2}K_3[1 + \cos(3\phi)] + \frac{1}{2}K_4[1 - \cos(4\phi)],$$

$$U_{\text{improper}} = K_{ijkl} (\psi - \psi_0)^2.$$





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/

Name	Modified
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Day1	34m ago
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conda-setup-guid...	34m ago
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conda-setup-guid...	34m ago
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environment.yml	34m ago
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Install_Wsl_Ubunt...	34m ago
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README.md	34m ago
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README.pdf	34m ago
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setup_conda_env.sh	34m ago
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Launcher



Python 3
(ipykernel)



Console



Python 3
(ipykernel)



Other



Terminal



Text File



Markdown File



Python File

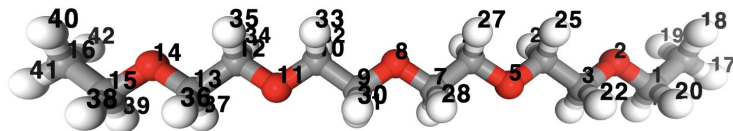


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Monomer structure



Monomer GAFF atom typing and partial charge parameters

CHARGE		0.00 (0)										
Formula:		H26 C12 O5										
ATOM	1	C	PEO	1	0.931	-0.097	-0.023	-0.097100		c3		
ATOM	2	C1	PEO	1	2.445	-0.109	0.007	0.132400		c3		
ATOM	3	O	PEO	1	2.928	0.653	-1.094	-0.428600		os		
ATOM	4	C2	PEO	1	4.355	0.677	-1.114	0.127400		c3		
ATOM	5	C3	PEO	1	4.829	1.500	-2.301	0.130400		c3		
ATOM	6	O1	PEO	1	6.254	1.515	-2.312	-0.430600		os		
ATOM	7	C4	PEO	1	6.750	2.278	-3.409	0.130400		c3		
ATOM	8	C5	PEO	1	8.269	2.261	-3.377	0.129400		c3		
ATOM	9	O2	PEO	1	8.762	3.024	-4.475	-0.429600		os		
ATOM	10	C6	PEO	1	10.186	3.036	-4.489	0.129400		c3		
ATOM	11	C7	PEO	1	10.660	3.858	-5.676	0.130400		c3		
ATOM	12	O3	PEO	1	12.084	3.872	-5.692	-0.430600		os		
ATOM	13	C8	PEO	1	12.575	4.634	-6.792	0.130400		c3		
ATOM	14	C9	PEO	1	14.095	4.618	-6.764	0.127400		c3		
ATOM	15	O4	PEO	1	14.598	5.379	-7.860	-0.428600		os		

Ambertools

1. antechamber

- Generates atom types (GAFF/GAFF2)
- Computes partial charges (AM1-BCC, RESP)
- Produces parameter files (`.mol2`, `.prepi`, `.frcmod`)

2. parmchk2

- Checks for missing parameters
- Generates a `.frcmod` file containing supplemental parameters

3. tleap

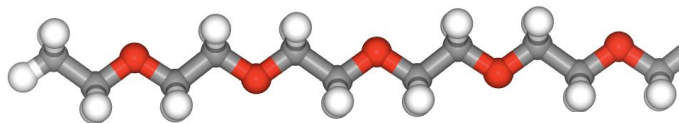
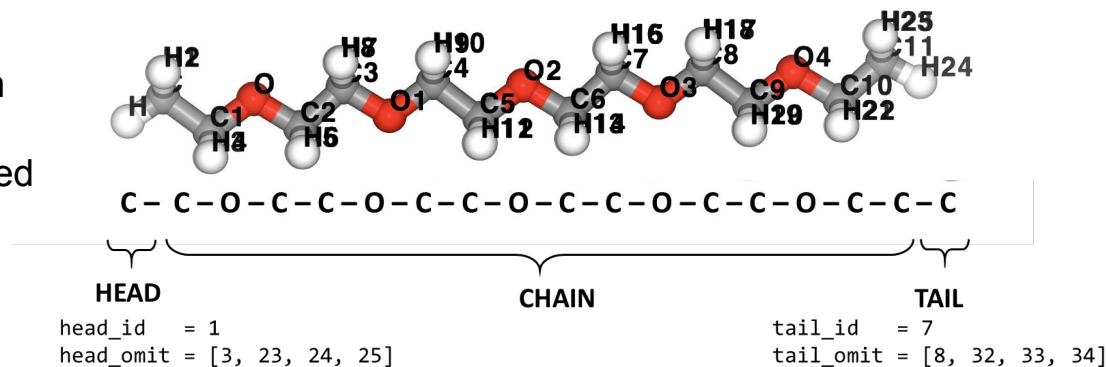
- Assembles the full system (molecule, solvent, ions)
- Writes AMBER `prmtop` and `inpcrd`

4. parmed

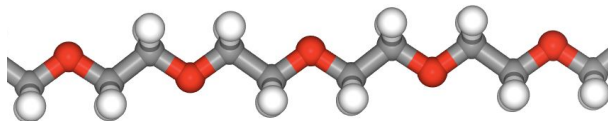
- Converts AMBER `prmtop` and `inpcrd` to GROMACS format

Polymer chain architecture - PEO

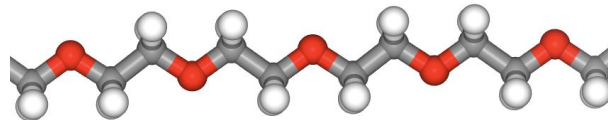
head_id is the Atom label where the Head end group is connected



Head (HTP)



Chain (PEO)

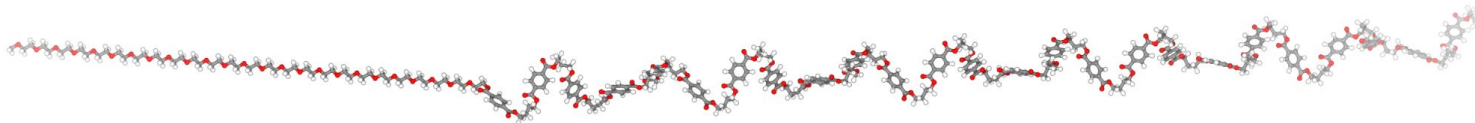


Tail (TPT)

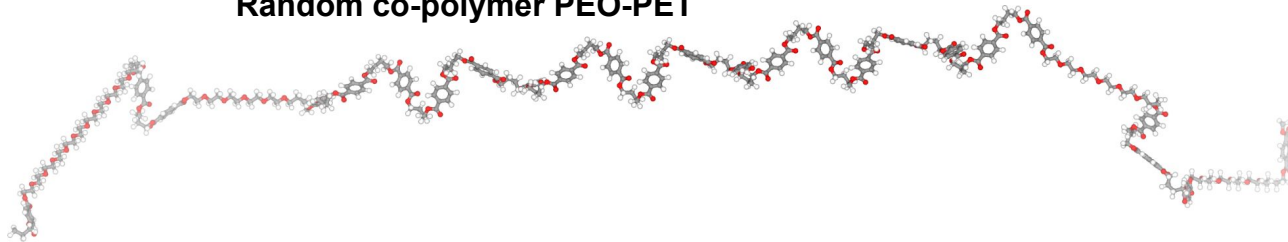
Polymer chain building with tleap

```
polymer_name = generate_polymer_tleap(  
    mol_names=["PEO", "PET"], # List of monomers to be present in polymer chain  
    n_mono_repeat=[5, 1], # List of monomers present in parameterised repeat unit  
    n_mono_pol=[25, 25], # Desired number monomers present in polymer chain  
    copolymer_type="random", # Type of polymer chain homo, block, random  
    head_group="PEO",  
    tail_group="PET"  
)  
  
print("Generated polymer:", polymer_name)
```

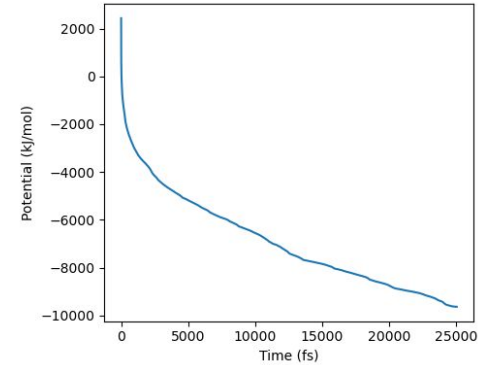
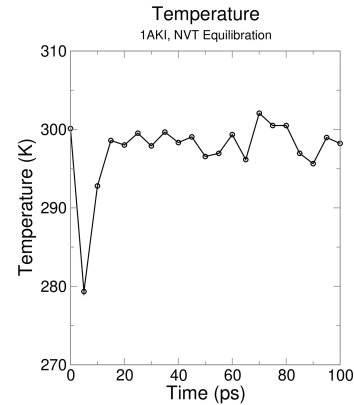
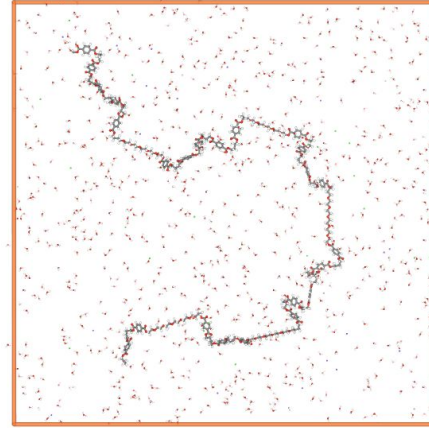
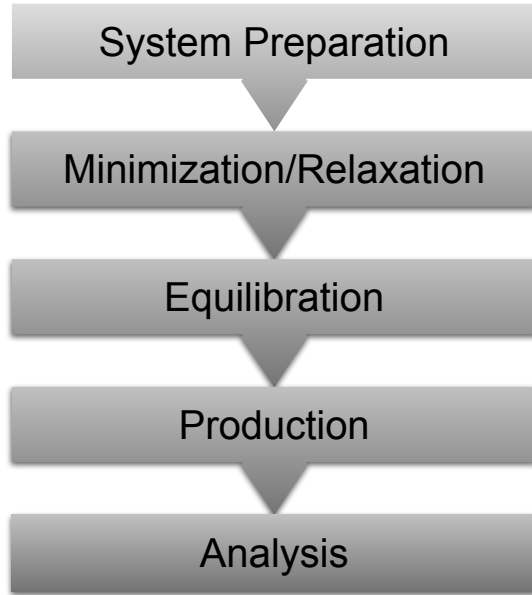
Block co-polymer PEO-PET



Random co-polymer PEO-PET



Main Stages of a MD simulation



GROMACS command-line

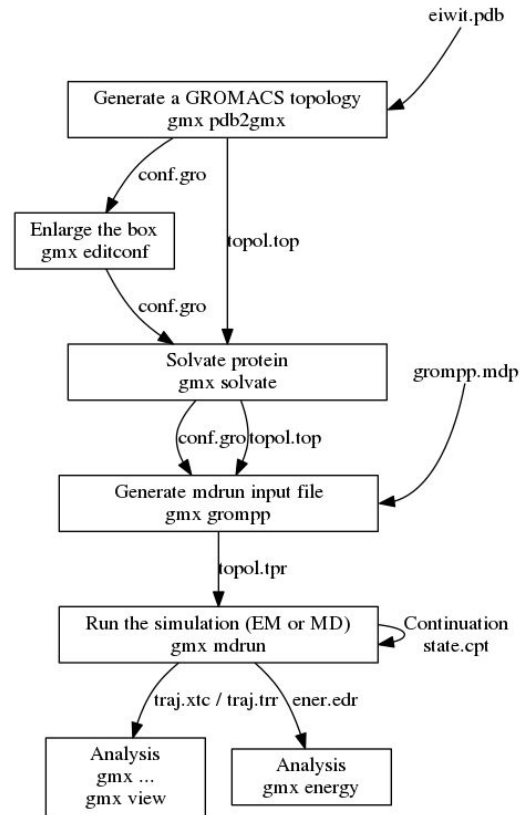
\$gmx **command** -options files/values/strings

pdb2gmx
editconf
genbox
solvate
grompp
mdrun
energy
....

```
OPTIONS
Options to specify input files:
-f      [<.gro/.g96/...>] (eiwit.pdb)
        Structure file: gro g96 pdb brk ent esp tpr
Options to specify output files:
-o      [<.gro/.g96/...>] (conf.gro)
        Structure file: gro g96 pdb brk ent esp
-p      [<.top>] (topol.top)
        Topology file
-i      [<.itp>] (posre.itp)
        Include file for topology
-n      [<.ndx>] (clean.ndx) (Opt.)
        Index file
-q      [<.gro/.g96/...>] (clean.pdb) (Opt.)
        Structure file: gro g96 pdb brk ent esp
Other options:
-chainsep <enum> (id_or_ter)
        Condition in PDB files when a new chain should be started (adding
        termini): id_or_ter, id_and_ter, ter, id, interactive
-merge <enum> (no)
        Merge multiple chains into a single [moleculetype]: no, all,
        interactive
-ff <string> (select)
        Force field, interactive by default. Use -h for information.
-water <enum> (select)
        Water model to use: select, none, spc, spce, tip3p, tip4p, tip5p,
        tips3p
```

\$gmx pdb2gmx -h

➔ To get help regarding the command's usage, description and options.



System Preparation

Minimization

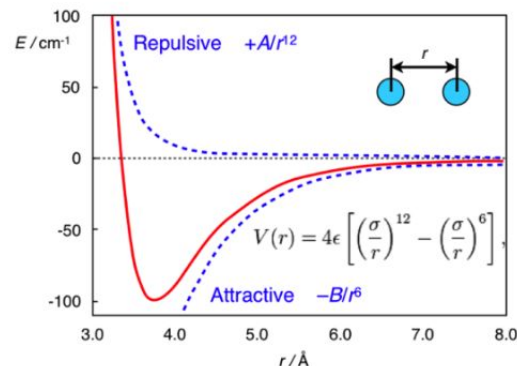
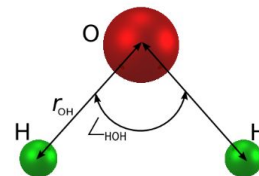
Equilibration

Production

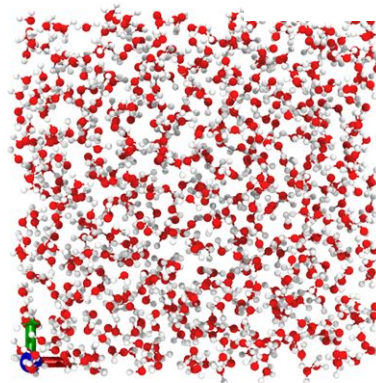
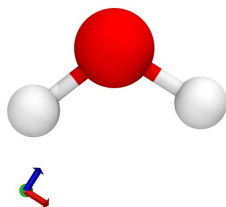
Analysis

Force field parameters – SPC/E water

potential	parameter	value
van der Waals	σ	3.166 Å
van der Waals	ϵ	0.650 kJ/mol
bond	r_{OH}	1.000 Å
angle	$\angle_{\text{HOH}}$	109.47°
electrostatic	q_{O}	-0.8476e
electrostatic	q_{H}	$ q_{\text{O}} /2$

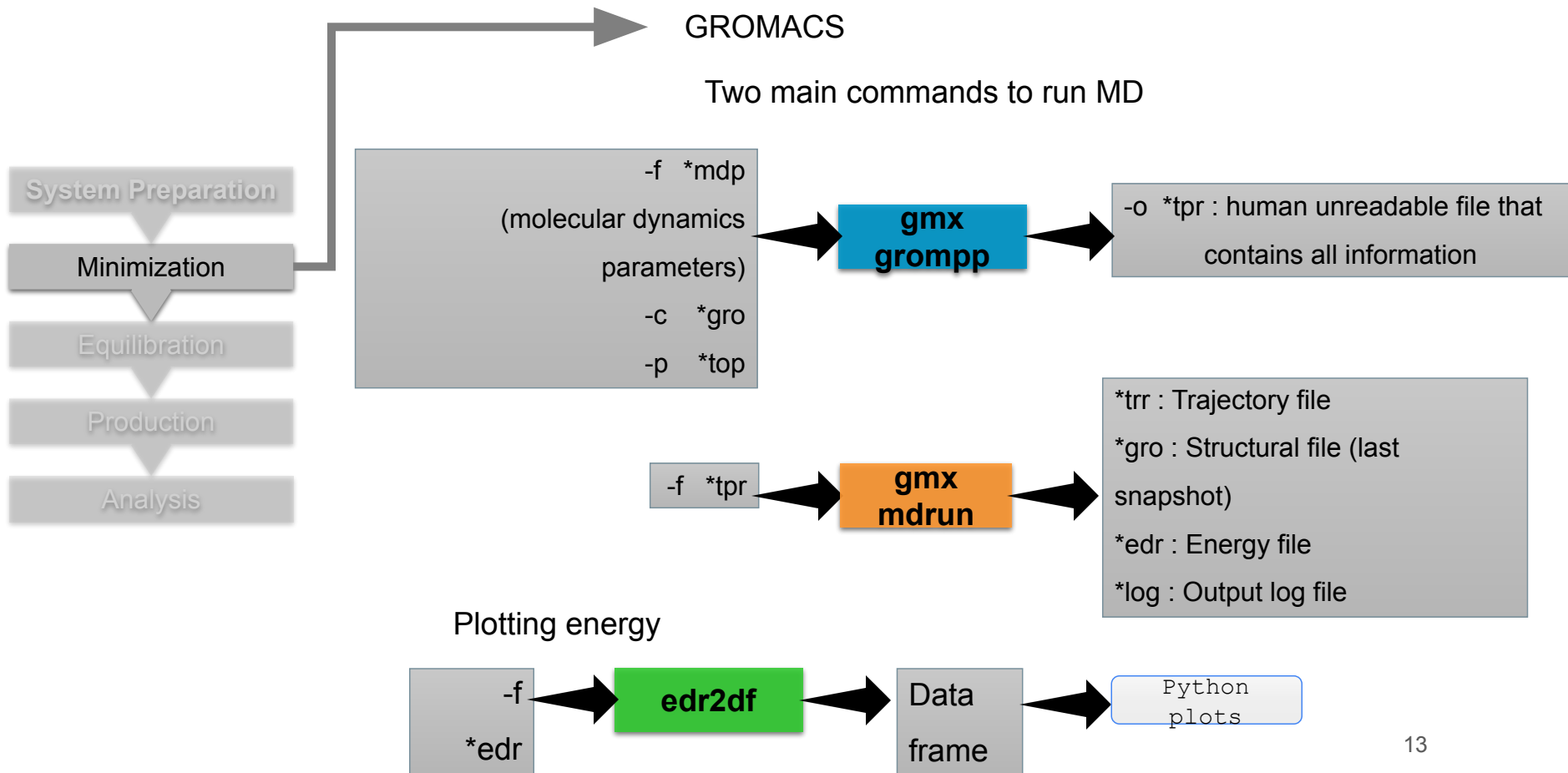


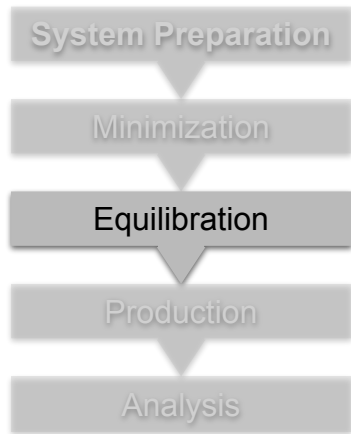
Initial Configuration



Check topol.top

```
[ molecules ]
; Compound      #mols
SOL              ?
SOL
```





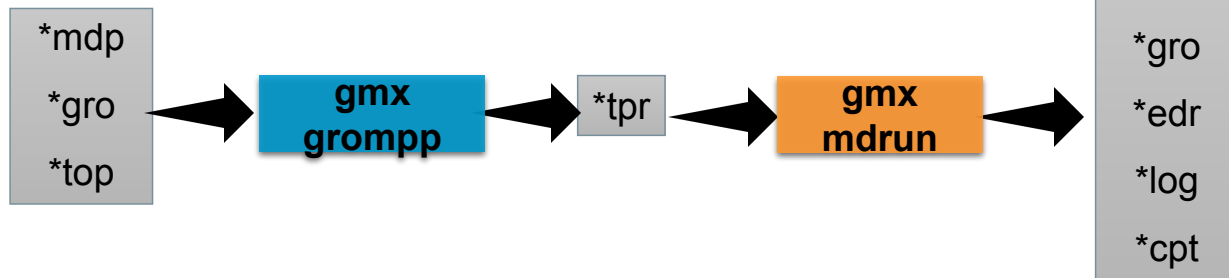
```

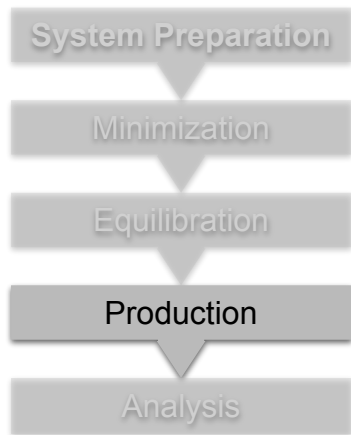
; Run parameters
integrator      = md
dt              = 0.002
nsteps         = 50000
  
```

```

; Setup the force field
constraints     = all-bonds
coulombtype     = PME
rcoulomb        = 1.0
vdwtype         = cut-off
rvdw            = 1.0
fourierspacing = 0.06
pme_order       = 4
ewald_rtol      = 1e-5
optimize_fft    = yes
  
```

Two main commands to run MD





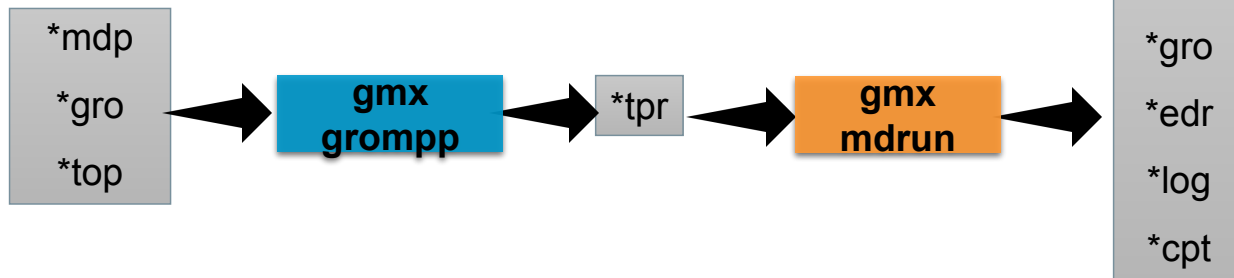
```

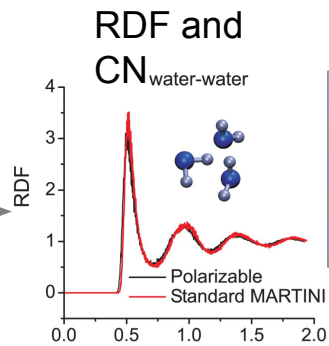
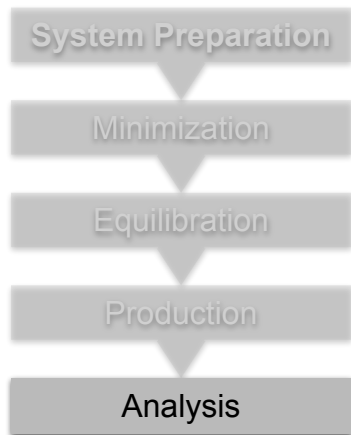
: Turning on thermostat
Tcoupl      = berendsen
tau_t       = 0.01
tc_grps     = System
ref_t       = 298
  
```

```

: Turn on barostat
Pcoupl      = berendsen
pcoupltype  = isotropic
tau_p       = 2.0
compressibility = 4.4e-5
ref_p       = 1.0
  
```

Two main commands to run MD





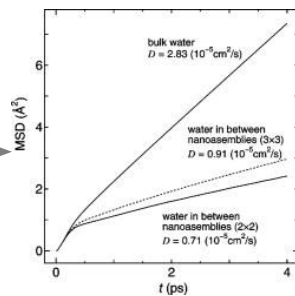
*trr
*tpr
*ndx

gmx rdf

df

Python
plot

MSD => D



*trr
*tpr
*nd
x

gmx msd

df

Python
plot